

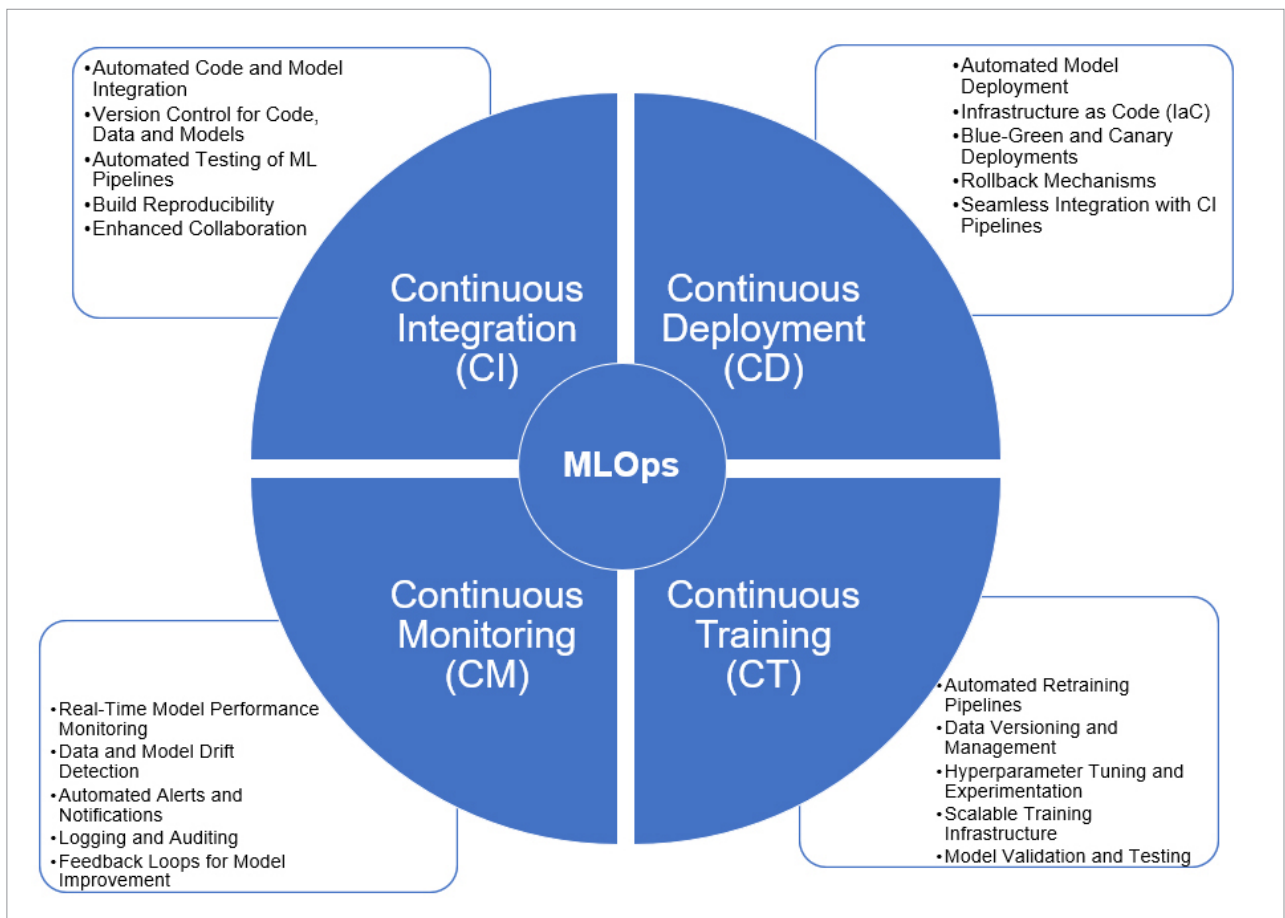


Figure 1: Principles of MLOps

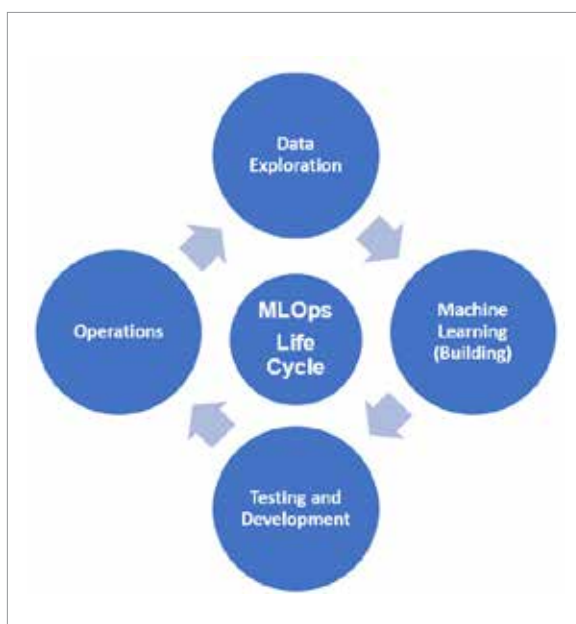


Figure 2: MLOps lifecycle

Data exploration: This initial phase occupies 30-40% of the project time. It involves collecting, transforming, and validating data to ensure it is ready for training.

Machine learning (building): This involves selecting the right algorithms and optimising model performance for the specific business use case.

Testing and development: Rigorous validation is done in this phase using techniques like cross-validation to ensure the model solves the problem without creating new ones.

Operations: This involves deploying the model as a service and monitoring its performance in a production environment.

MLOps framework for enterprise adoption

The MLOps framework is designed to bridge the gap between model development and production. It focuses on automation, monitoring, and standardising the ML lifecycle. The steps involved are:

Business case: In this step, the usage of ML and how it should be leveraged to solve the business problem is clearly articulated.

Data collection: Raw data is identified and gathered from all relevant sources as part of this step – for example, EHR data, lab results,